

Xiao Zhang

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ABOUT ME

I am a Ph.D. student at UT Austin advised by Prof. Daehyeok Kim, working at the intersection of networked/distributed systems and AI infrastructure. My work spans datacenter networks, 5G and edge computing, and cross-layer telemetry, with a current focus on making AI workloads predictable and efficient—from the network fabric up to GPU resources. I build practical, measurement-driven systems that bridge real-world deployment with the infrastructure needs of modern AI.

EXPERIENCE

Google Cloud

Software Engineering Intern

Sunnyvale, CA

June 2026 - Present

- Developing network telemetry for LLM workload analysis

University of Texas at Austin (with Prof. Daehyeok Kim)

Ph.D. Student

Austin, TX

Sept. 2024 - Present

- Designed and built a private 5G testbed with GPU-accelerated edge servers for studying AI inference latency over cellular networks
- Conducted large-scale measurements using AWS Wavelength Zones and Verizon 5G to identify latency variability in edge AI applications
- Identified key performance bottlenecks in wireless scheduling and GPU resource contention; currently developing cross-layer resource management techniques for predictable edge inference latency

University of Pennsylvania (with Prof. Vincent Liu)

Visiting Student

Philadelphia, USA

July. 2023 - Present

- *Beaver*: Enabling Practical Distributed Snapshots Exploiting Software Load Balancers

Shanghai Jiao Tong University (with Prof. Shizhen Zhao)

Master Student

Shanghai, China

Sep. 2021 - July. 2023

- *Flattened Clos Plus (FC+)*: Near-optimal topology-routing co-design free of deadlocks for RoCE-based expander networks
- *Flattened Clos (FC)*: Deadlock-free topology-routing co-design for RoCE-based expander networks

HONORS AND AWARDS

2025 **Amazon AI PhD Fellowship**, Awardee

Austin, TX

2021 **Outstanding Graduate of Shanghai**, Awardee

Shanghai

2020 **Liu Yongling Scholarship**, Awardee

Shanghai

EDUCATION

University of Texas at Austin

Ph.D. in Computer Science

Austin, Texas

Sept. 2024 - Present

- Advisor: Prof. Daehyeok Kim
- Research focus: Networked & distributed systems, AI infrastructure, Edge computing

Shanghai Jiao Tong University
M.E. in Communication Engineering

Shanghai, China
Sept. 2021 - May. 2024

- Thesis title: FC+: Near-optimal Deadlock-free Expander Data Center Networks

Shanghai Jiao Tong University
B.E. in Information Engineering

Shanghai, China
Sept. 2017 - June. 2021

- Thesis title: Design of Robust and Efficient Edge Server Placement and Server Scheduling Policies

PUBLICATION

- **Xiao Zhang**, Daehyeok Kim. [Enabling SLO-Aware 5G Multi-Access Edge Computing with SMEC](#). In Proceedings of 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), May 2026.
- Liangcheng Yu, **Xiao Zhang**, Haoran Zhang, John Sonchack, Dan Ports, Vincent Liu. [Beaver: Practical Partial Snapshots for Distributed Cloud Services](#). In Proceedings of 18th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), July 2024.
- **Xiao Zhang**, Peirui Cao, Yongxi Lyu, Qizhou Zhang, Shizhen Zhao, Xinbing Wang, Chenghu Zhou. [FC+: Near-optimal Deadlock-free Expander Data Center Networks](#). In Proceedings of 21st IEEE International Symposium on Parallel and Distributed Processing with Applications (**ISPA**), December 2023.
- Shizhen Zhao*, Qizhou Zhang*, Peirui Cao, **Xiao Zhang**, Xinbing Wang, Chenghu Zhou. [Flattened Clos: Designing High-performance Deadlock-free Expander Data Center Networks Using Graph Contraction](#). In Proceedings of 20th USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), 2023.
- Shizhen Zhao*, **Xiao Zhang***, Peirui Cao, Xinbing Wang. [Design of Robust and Efficient Edge Server Placement and Server Scheduling Policies](#). In Proceedings of IEEE/ACM 29th International Workshop on Quality of Service (**IWQoS**), 2021.

SKILLS

Programming	C/C++, Python, Verilog, VHDL, Matlab
Tools	eBPF, DPDK, FPGA, Network Simulator 3 (NS-3), AWS
Languages	English, Chinese